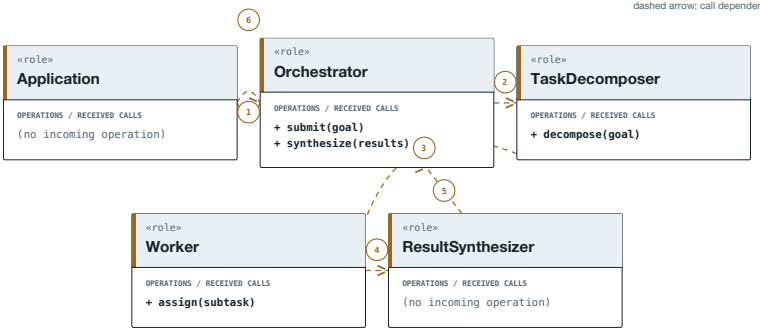
# W04 Orchestrator-workers

**Workflow / UML CLASS VIEW / ANALYTICAL MODEL**

Generate and delegate subtasks dynamically.

**UML CLASS VIEW**

dashed arrow: call dependency · brown: return/failure · analytical, not an observed trace




**POINTCUT / PATTERN INTERFACE**

| | |
|---|---|
| CONTEXT | A lead agent can generate tasks rather than call a fixed pipeline. |
| TRIGGER | The orchestrator decomposes a goal or dispatches a subtask. |
| STRUCTURE | Task plan, worker selection, delegation envelope, and synthesis step. |
| EFFECT | Synthesized output accounts for generated task results and delegated authority. |

**CONTRACT / REQUIRED BEHAVIOR**

| | |
|---|---|
| REQUIRES | Workers advertise capabilities and share a task/result protocol. |
| ENSURES | Every dispatched task has a result or failure and the final answer links them. |
| INVARIANT | Dynamic dispatch cannot create unbounded uncounted worker expansion. |
| VIOLATION | Unsplittable, failed, or duplicate work is reflected in synthesis. |

**RELATIONSHIPS / ANALYTICAL CALLS**

number · caller → callee · message

1 Application → Orchestrator submit(goal)
2 Orchestrator → TaskDecomposer decompose(goal)
3 TaskDecomposer → Worker assign(subtask)
4 Worker → ResultSynthesizer return(result)
5 ResultSynthesizer → Orchestrator synthesize(results)
6 Orchestrator → Application return(answer)

**VARIATION / TEST / NEGATIVE CONTROL**

| | |
|---|---|
| VARIATION | Planner-executor, manager-worker, sub-agent registry, and dynamic graph. |
| TEST_OBLIGATION | Test decomposition quality, worker failure, duplicate task, and recursion cap. |
| NEGATIVE_CONTROL | A fixed sequence of sub-agents is W01 rather than dynamic W04. |